Topic 13: Introduction to Probability

02-680: Essentials of Mathematics and Statistics

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Probability is the mathematical language for quantifying uncertainty

1 Basics

The ***sample space*** (Ω), sometimes called the universe, is the set of all possible outcomes of an experiment (or random process). The outcomes are *mutually exclusive*: exactly one outcome happens each time, and together they cover everything that could happen. (This Ω is not the Ω from asymptotics; same letter, different object.)

$$\Omega_{coin} = \{\texttt{Heads}, \texttt{Tails}\}$$

$$\Omega_{twocoin} = \Omega_{coin} \times \Omega_{coin} = \{\langle \texttt{Heads}, \texttt{Heads}\rangle, \langle \texttt{Heads}, \texttt{Tails}\rangle, \langle \texttt{Tails}, \texttt{Heads}\rangle, \langle \texttt{Tails}, \texttt{Tails}\rangle\}$$

A sample ***outcome***, or atomic element, $\omega \in \Omega$ is one of the possible things that can happen.

An ***event*** is a ***subset*** of the sample space: a group of outcomes that we bundle together because we care about them as a group. Outcomes are the ***elements*** of Ω; events are the ***subsets*** of Ω. For instance, the event that the first of two coin tosses is heads:

$$A_{firstheads} = \{\langle \texttt{Heads}, \texttt{Heads}\rangle, \langle \texttt{Heads}, \texttt{Tails}\rangle\} \subseteq \Omega_{twocoin}$$

We say the ***probability*** $p(A)$ of an event A is the fraction of all outcomes that A covers. (For now we assume all outcomes are equally likely; the axioms below describe what p must satisfy in general.) So in the example above $p(A_{firstheads}) = \frac{1}{2}$. Notice that p takes an ***event*** (a set), not an outcome. When we write $p(\texttt{Heads})$ we really mean $p(\{\texttt{Heads}\})$, the event containing only that outcome.

Event space. The ***event space***, $\mathcal{A}$, is the set of all events we are allowed to ask about. That makes it a set of sets. (We use the calligraphic $\mathcal{A}$ for the set of events and a plain A for a single event.) For a single coin, $\Omega_{coin} = \{\texttt{Heads}, \texttt{Tails}\}$ has two outcomes, but four events:

$$\mathcal{A}_{coin} = \Big\{ \emptyset,\ \ \{\texttt{Heads}\},\ \ \{\texttt{Tails}\},\ \ \{\texttt{Heads}, \texttt{Tails}\} \Big\}$$

("nothing happens," "heads," "tails," and "heads or tails"). For discrete sample spaces (such as coin flips) we can typically take $\mathcal{A}$ to be the power set of Ω, so $|\mathcal{A}| = 2^{|\Omega|}$. The two-coin sample space has $|\Omega_{twocoin}| = 4$ outcomes and so $2^4 = 16$ events. In continuous spaces it is typically thought of as the Borel field of Ω (details of this are beyond the scope of the class for now).

	sample space Ω	event space $\mathcal{A}$
contains	outcomes (what can happen)	events (what we can ask about)
its members are	elements $\omega \in \Omega$	subsets $A \subseteq \Omega$, so $A \in \mathcal{A}$
one coin	2 outcomes	4 events
two coins	4 outcomes	16 events

Think of it this way

Ω is the list of everything that could be in the freezer: each tissue sample is an outcome. $\mathcal{A}$ is the list of every yes/no question you could ask about the sample you pulled, such as "is it high expression?" Each question is the ***set*** of outcomes for which the answer is yes, so each question is an event.

Why the event space has rules. If we can ask whether A happened, we should also be able to ask whether it did ***not*** happen, and whether A or B happened:

(1) $A \in \mathcal{A} \to \overline{A} \in \mathcal{A}$ (we can ask "not A"; the ***complement*** $\overline{A} = \Omega \setminus A$ is everything in Ω that is not in A)

(2) $A_1, A_2, ... \in \mathcal{A} \to \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$ (we can ask "A_1 or A_2 or ...")

These are exactly what the axioms below need: Axiom 3 adds probabilities of unions, and the complement rule $p(\overline{A}) = 1 - p(A)$ only makes sense if $\overline{A}$ is an event.

1.1 Three Axioms

Axiom 1: Non-negative Probability. For any event A, $p(A) \geq 0$.

That is, we cannot have a negative probability (zero is allowed). In the previous example if $A_{threetails}$ is the event that you get 3 tails when a coin is tossed twice, then $A_{threetails} = \emptyset$ (it can't happen), so $p(A_{threetails}) = 0$ but this can't be negative.

Axiom 2: Total Probability. For any sample space Ω, $p(\Omega) = 1$. That is, the probability of something in the sample space happening is 1. So for the example above the probability of a single coin flip being either `Heads` or `Tails` is 1, (as defined) there are no other possible outcomes.

Axiom 3: Disjoint Events. For *pairwise disjoint* events $A_1, A_2, ..., A_n$ (no two of them share an outcome),

$$p(A_1 \cup A_2 \cup ... \cup A_n) = p(A_1) + p(A_2) + ... + p(A_n).$$

So in a single coin flip if A_{heads} and A_{tails} are the events of a single coin flip being heads and tails respectively, we can see the events are disjoint (they don't share any outcomes) so

$$p(A_{heads} \cup A_{tails}) = p(A_{heads}) + p(A_{tails}) = \frac{1}{2} + \frac{1}{2} = 1.$$

As a counter example lets define

$$A_{lastheads} = \{\langle \texttt{Heads}, \texttt{Heads} \rangle, \langle \texttt{Tails}, \texttt{Heads} \rangle\} \subset \Omega_{twocoin}.$$

We can see that

$$p(A_{firstheads} \cup A_{lastheads}) \neq p(A_{firstheads}) + p(A_{lastheads})$$

because $A_{firstheads} \cap A_{lastheads} = \{\langle \texttt{Heads}, \texttt{Heads} \rangle\} \neq \emptyset$ thus they are not disjoint. In fact the union has only 3 outcomes, so $p(A_{firstheads} \cup A_{lastheads}) = \frac{3}{4}$, but $p(A_{firstheads}) + p(A_{lastheads}) = \frac{1}{2} + \frac{1}{2} = 1$: the outcome $\langle \texttt{Heads}, \texttt{Heads} \rangle$ was counted twice.

1.2 Example Consequences of the Three Axioms

1.2.1 $p(\emptyset) = 0$

$\Omega \cup \emptyset = \Omega$.

Since Ω is disjoint from $\emptyset$, we know $p(\Omega \cup \emptyset) = p(\Omega) + p(\emptyset)$.

Put these two together we see that $p(\Omega) = p(\Omega) + p(\emptyset)$.

Since $p(\Omega) = 1$, then $1 = 1 + p(\emptyset)$, and thus $p(\emptyset) = 0$.

1.2.2 $p(\overline{A}) = 1 - p(A)$

We know by definition that $\overline{A} = \Omega \setminus A$.

From Axiom 3, since A and $\overline{A}$ are disjoint, we know $p(A \cup \overline{A}) = p(A) + p(\overline{A})$.

But we also know that $A \cup \overline{A} = \Omega$.

So that means $p(A) + p(\overline{A}) = p(\Omega) = 1$. With some algebra we find the original statement.

1.2.3 $0 \leq p(A) \leq 1$.

From axiom 1: $0 \leq p(A)$.

For the other side, the previous result gives $p(A) = 1 - p(\overline{A})$, and $p(\overline{A}) \geq 0$ by axiom 1, so $p(A) \leq 1 - 0 = 1$.

1.2.4 If $A \subseteq B$ then $p(A) \leq p(B)$.

$$\begin{aligned} p(B) &= p(A \cup (B \setminus A)) && B = A \cup (B \setminus A) \\ &= p(A) + p(B \setminus A) && A \text{ and } B \setminus A \text{ are disjoint (Axiom 3)} \\ &\geq p(A) && p(B \setminus A) \geq 0 \text{ (Axiom 1)} \end{aligned}$$

2 Perspectives

For a coin toss, there are two common ways of explaining what $p(\texttt{Heads}) = \frac{1}{2}$ means:

Frequentists. Probability is a long-run frequency. You flip a coin 100 times, you get `Heads` approximately 50 of those tosses, and the more you flip the closer the fraction of `Heads` gets to $\frac{1}{2}$.

Bayesian. Probability is a degree of belief. You *believe* you will get heads 50% of the times you flip the coin, and you update that belief as you see evidence (for instance, a coin that keeps landing `Tails`).

Both views follow the same three axioms and the same rules derived from them, so everything in this topic holds either way; they differ in what the number $p(A)$ is said to ***mean***.

Useful References

Wasserman. "All of Statistics: A Concise Course in Statistical Inference" §§1.2–1.3
Degroot and Schervish. "Probability and Statistics" §§1.2–1.5